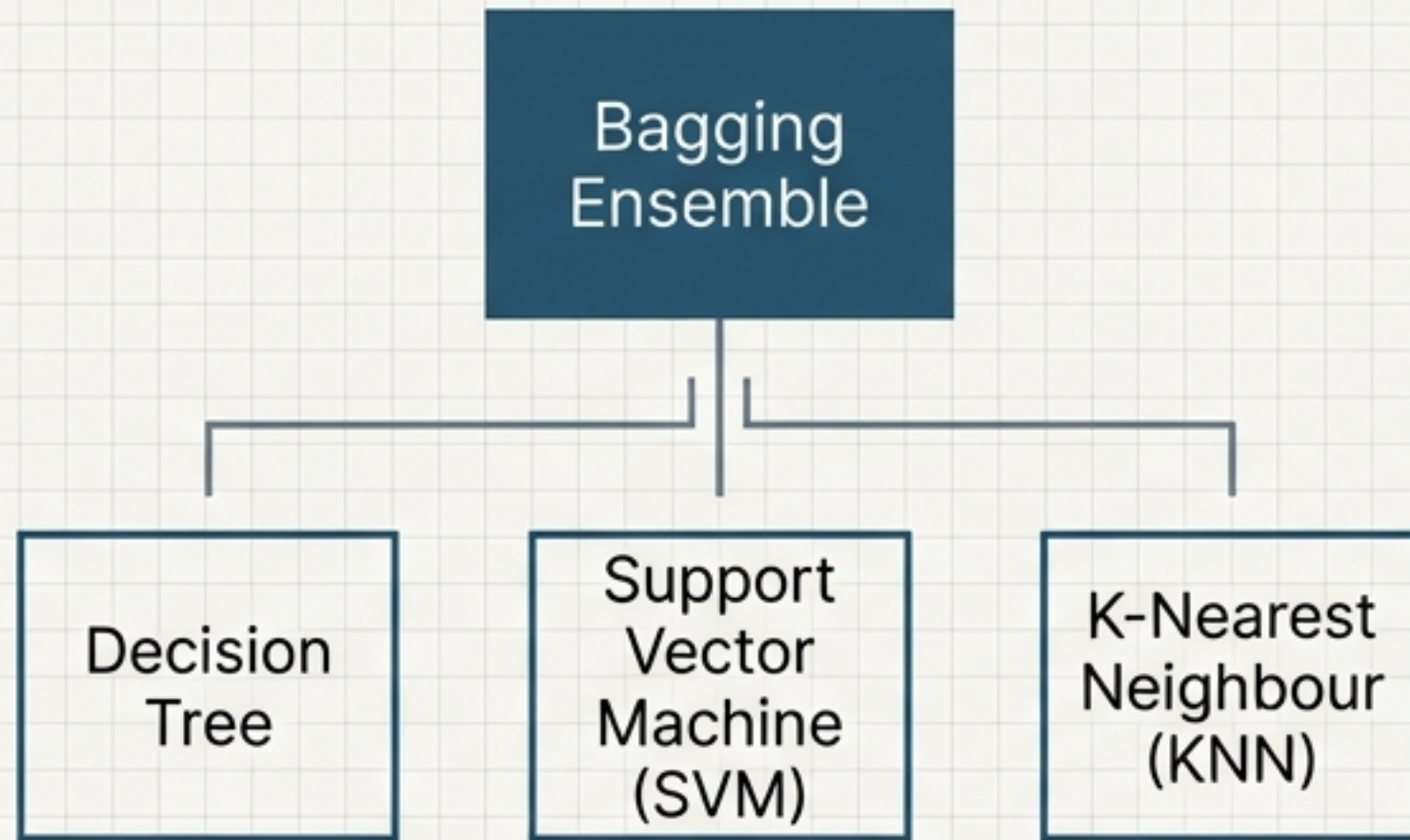


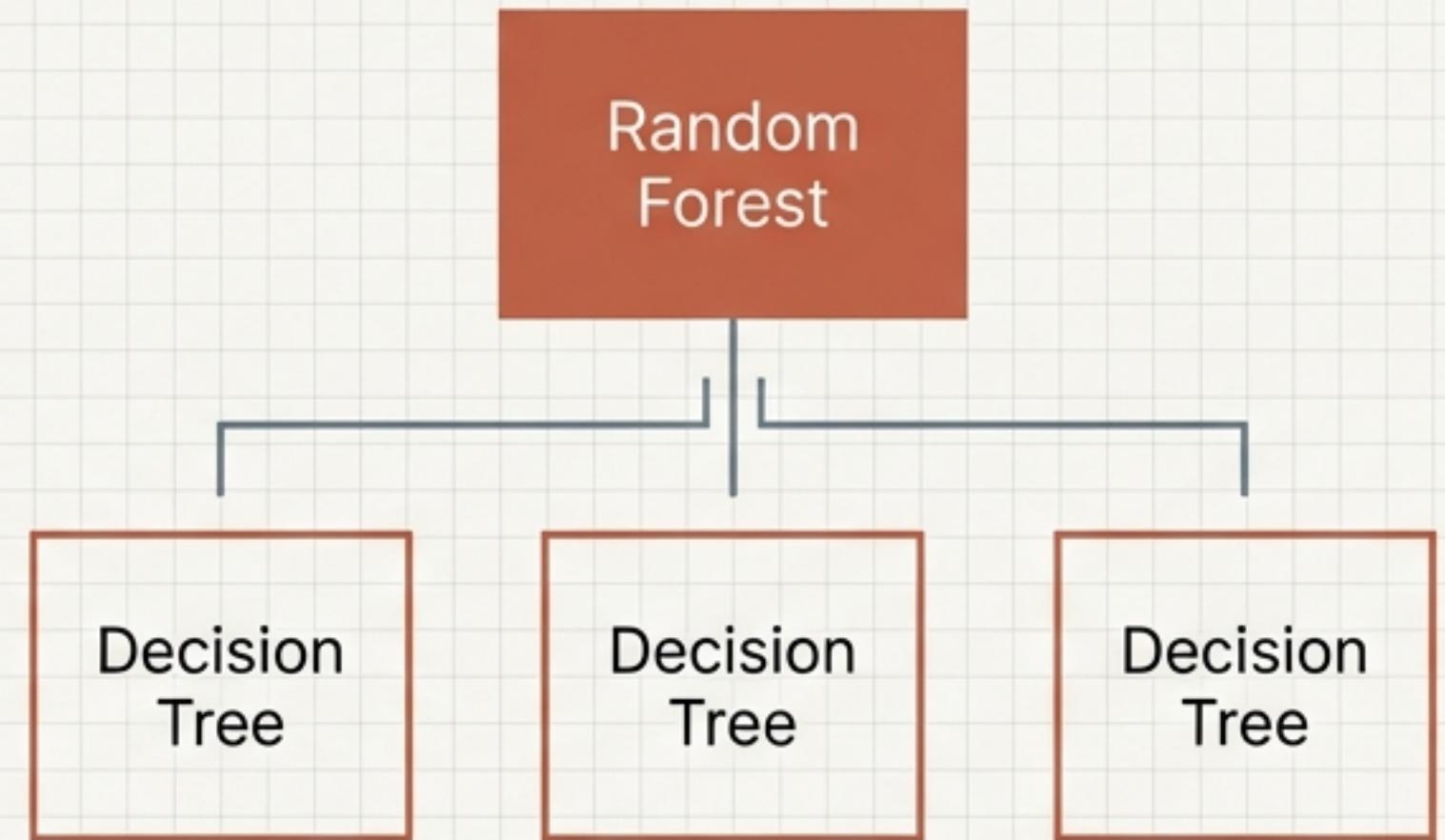
Bagging(Base = Decision Trees) $\neq$ Random Forest

Deconstructing the definitive architectural difference between two foundational ensemble methods.

Algorithmic Flexibility vs. Structural Rigidity



A generalised ensemble technique. The base estimator can be any algorithm.



A highly specialised ensemble. The base estimator is strictly restricted to Decision Trees.

The Scikit-Learn API Reflects the Architecture

Algorithm 1

`sklearn.ensemble.BaggingClassifier`

`estimator=None`

Defaults to a Decision Tree, but accepts any structural estimator.

Algorithm 2

`sklearn.ensemble.RandomForestClassifier`

~~`estimator`~~

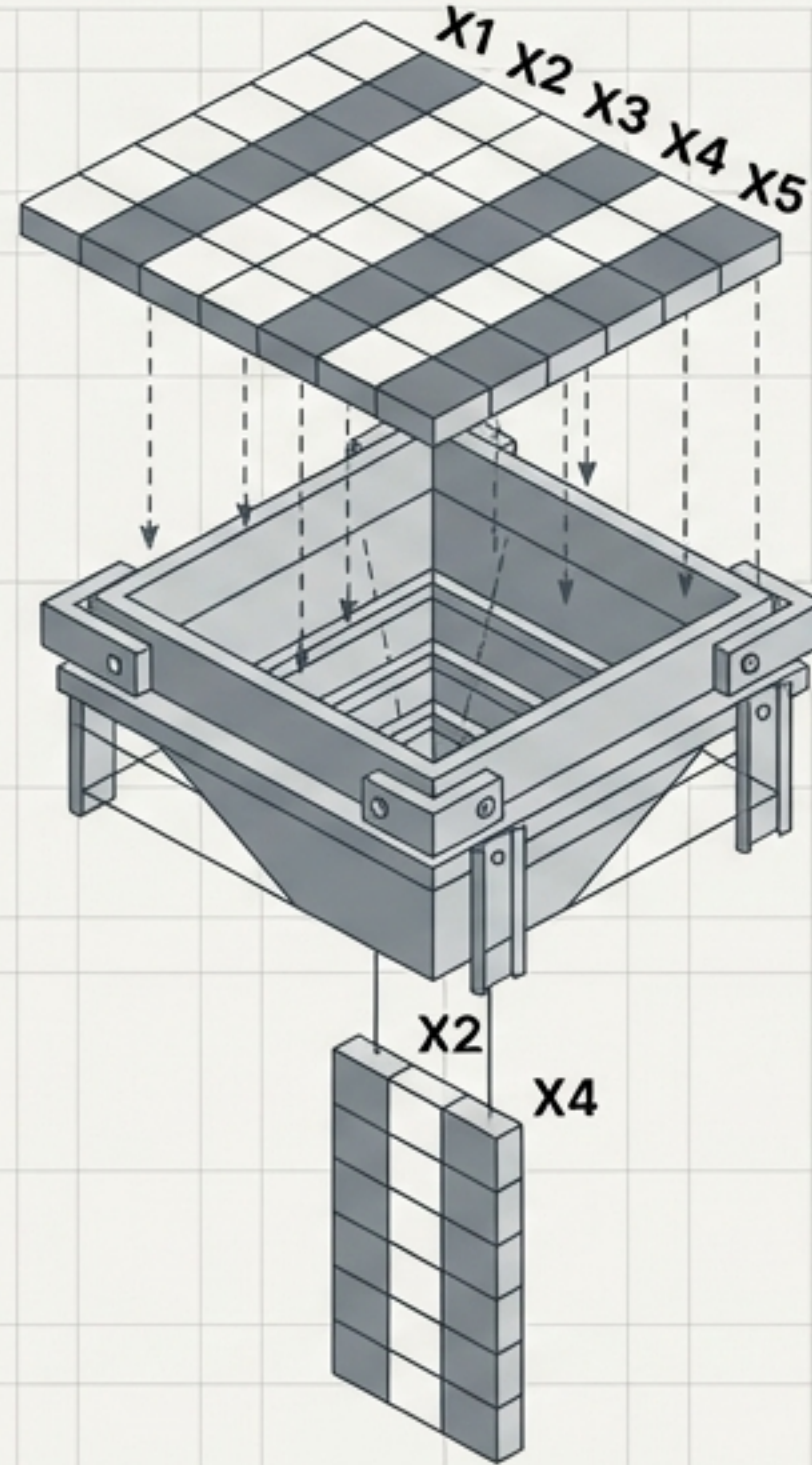
There is no base estimator parameter. The algorithm is hardcoded to use trees.



The Interview Trap: If you configure a Bagging classifier and explicitly set its base models to Decision Trees, have you just built a Random Forest?

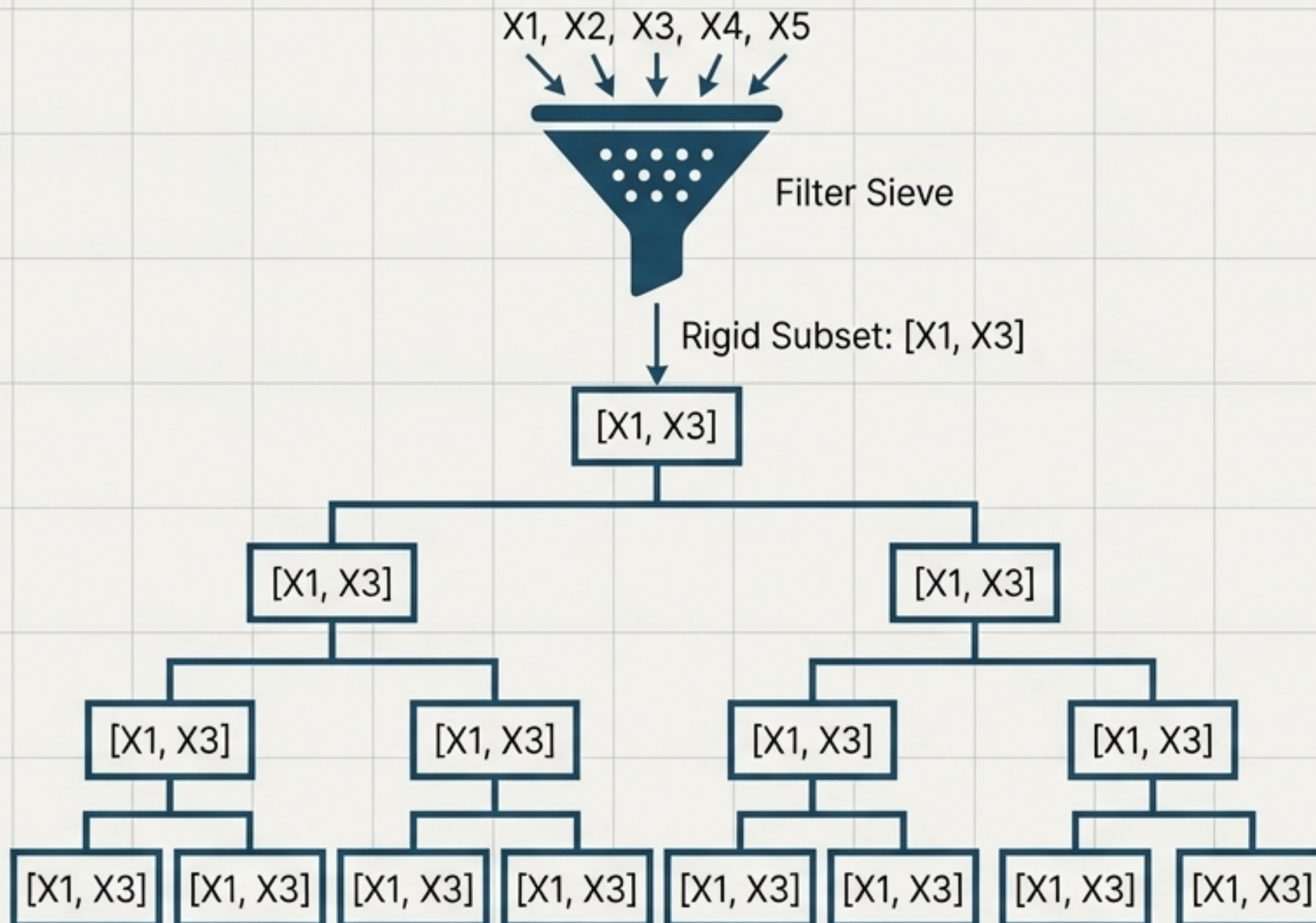
Most candidates say yes. The structural reality is no.

The Hidden Mechanism: Column Sampling Scope



Both algorithms use Column Sampling (passing only a random subset of features to the models). The critical architectural difference lies in when and where this sampling occurs within the tree structure.

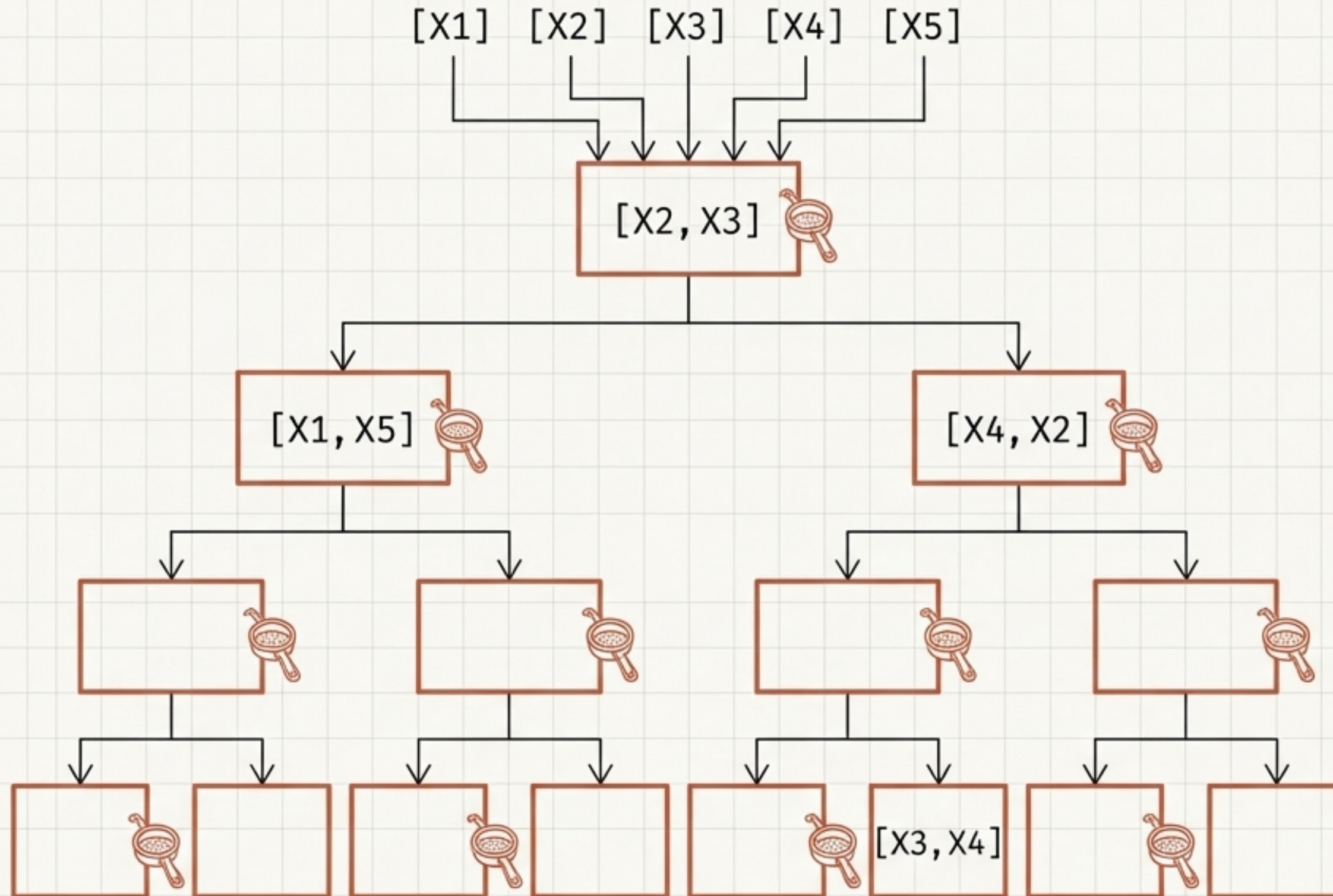
Bagging Implements Tree-Level Sampling



Features are randomly sampled exactly once, prior to tree construction.

The entire tree is forced to build itself using only that restricted subset of columns.

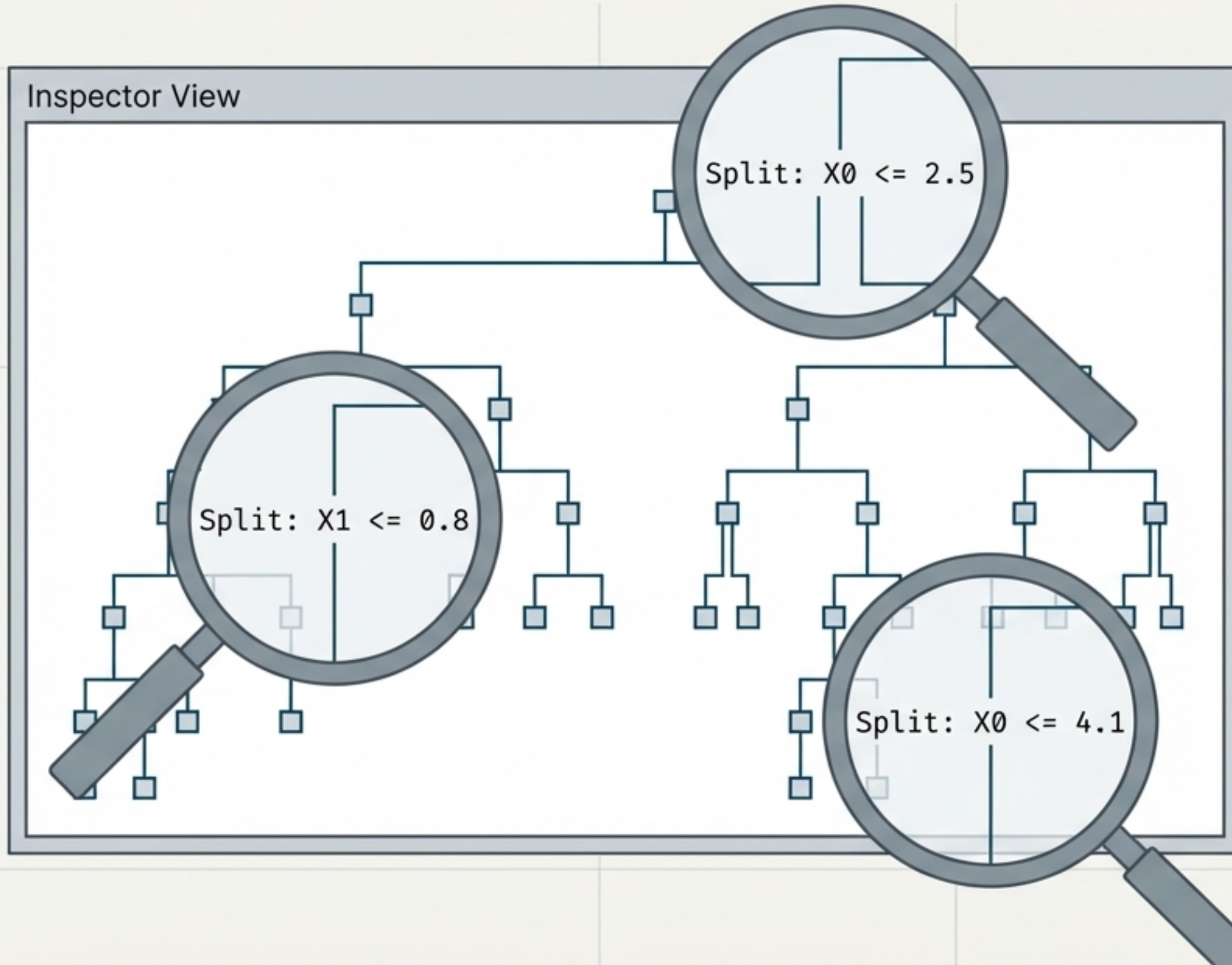
Random Forest Implements Node-Level Sampling



Features are dynamically sampled at every individual split.

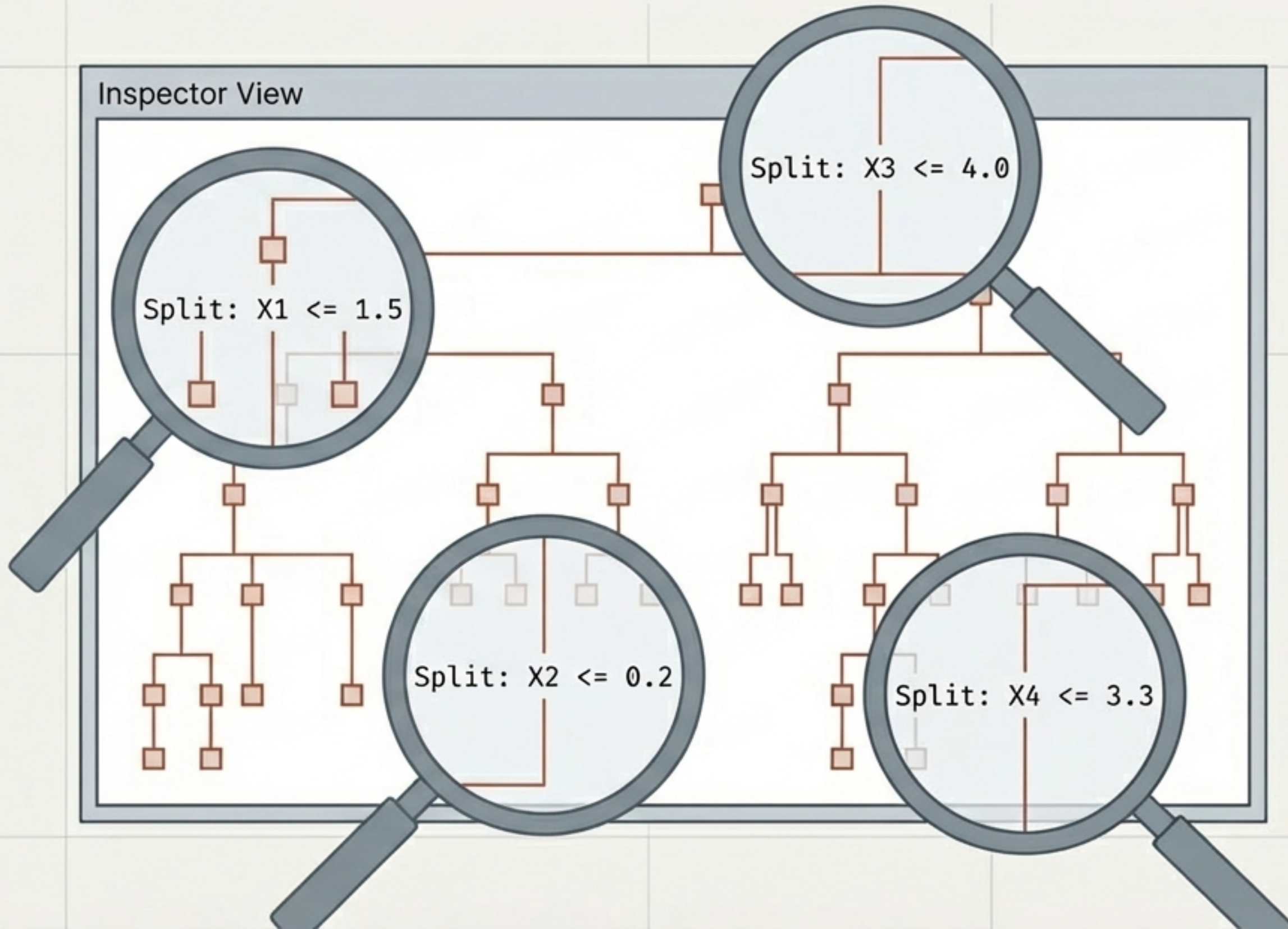
The tree constantly introduces new random feature subsets as it grows deeper.

Empirical Proof: Inside the Bagging Blueprint



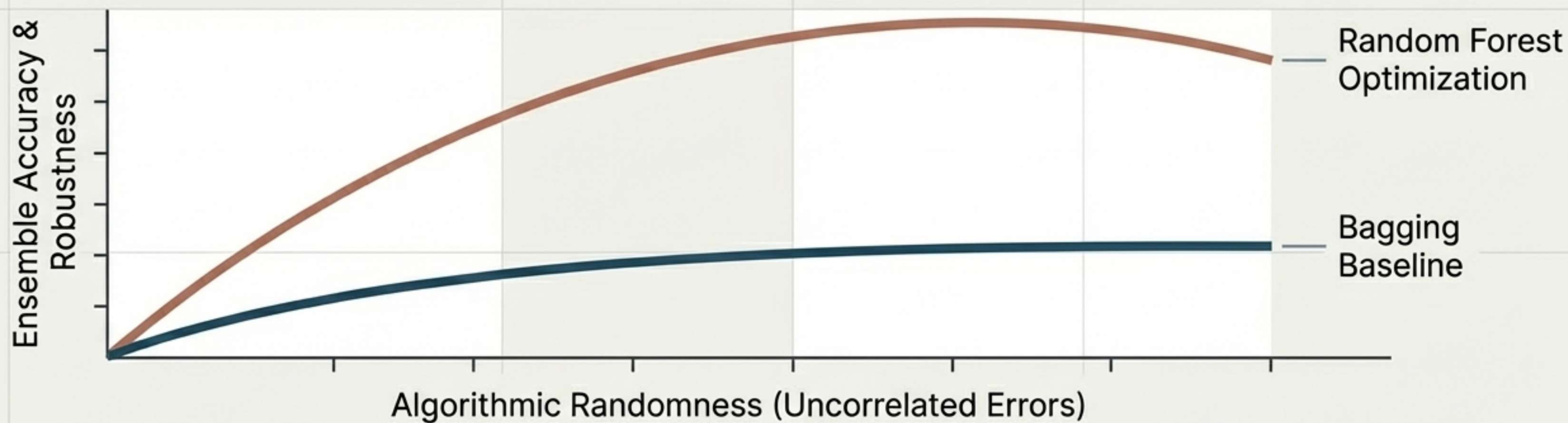
When plotting a Bagging tree, the structural rigidity is visible. Despite having 5 available features, every split across the entire tree relies exclusively on the initially sampled $[X_0, X_1]$.

Empirical Proof: Inside the Random Forest Blueprint



Plotting a Random Forest reveals dynamic injection. Because sampling occurs at the node level, almost all available features eventually participate in the tree's construction.

The Performance Quotient: Randomness = Accuracy



Ensemble power relies on base models making uncorrelated errors. By forcing node-level sampling, Random Forest injects drastically more randomness into the system. High variance among base estimators (provided individual accuracy >50%) yields a demonstrably stronger, more robust aggregated model.

The Definitive Diagnostic Matrix

	Bagging	Random Forest
Base Estimator	Any Algorithm	Strictly Decision Trees
Scikit-Learn API	estimator parameter exposed	Hardcoded; parameter omitted
Column Sampling Scope	Macro (Tree-Level)	Micro (Node-Level)
Randomness Quotient	Moderate	High (Maximises un-correlated errors)